

A Non-Archimedean Operator Algebra with Type-II Baer Reduction

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August 9, 2026

Abstract

Kochubei asked for a non-Archimedean operator algebra whose reduction is a type-II Baer ring. We give a general equal-characteristic realization principle: every unital algebra R over a field k is the reduction of a norm-closed unital operator algebra on a space $c_0(J, k((t)))$. Choose a Hamel basis of R containing 1, put the Gauss norm on $k((t)) \otimes_k R$, complete it, and use the left regular representation. The trivial norm on the coefficient field makes the Gauss norm submultiplicative, while evaluation at 1 makes the regular representation isometric. A c_0 -family has only finitely many coefficients with nonzero residue, so completion introduces no new residue elements. Applying the construction to von Neumann's algebraic continuous factor $\mathcal{M}_k$, a type-II_f regular self-injective ring, produces the required type-II Baer reduction.

1 Source problem and scope

Let K be a complete nontrivially valued non-Archimedean field, let $\mathcal{O}_K$ be its valuation ring, $\mathfrak{m}_K$ its maximal ideal, and $\tilde{K} = \mathcal{O}_K/\mathfrak{m}_K$ its residue field. For a norm-closed unital algebra $\mathfrak{A} \subseteq \mathcal{L}(c_0(J, K))$, write

$$\mathfrak{A}_1 = \{T \in \mathfrak{A} : \|T\| \leq 1\}, \quad \mathfrak{A}_0 = \{T \in \mathfrak{A} : \|T\| < 1\}, \quad \tilde{\mathfrak{A}} = \mathfrak{A}_1/\mathfrak{A}_0.$$

Kochubei calls $\mathfrak{A}$ an operator algebra with Baer reduction when $\tilde{\mathfrak{A}}$ is a Baer ring [3]. At the end of Section 4, on page 7, he states:

“The problem of constructing non-Archimedean operator algebras with type II Baer reductions remains open.”

The result below answers the literal existence problem in the general class of complete non-Archimedean fields used in the paper's definition. It gives an equal-characteristic example and does not assert that the example is a group algebra. Section 4 of the source paper begins with the temporary convention $K = \mathbb{C}_p$. If the final sentence is instead read as a problem specifically over that fixed mixed-characteristic field, our construction is a substantial partial answer but does not settle that narrower interpretation; see Section 5.

2 Proof intuition

The key point is that equal characteristic provides a coefficient field. Take $K = k((t))$, so that k embeds in the valuation ring and maps isomorphically onto the residue field. If E is a Hamel basis of an arbitrary k -algebra R , every product of two basis elements is a finite k -linear combination of basis elements. Since every nonzero scalar in k has t -adic norm one, the resulting structure constants

is a finite direct sum of full matrix algebras over $\mathcal{F}_p$. Therefore $\mathcal{F}_p[G]$ is a type I Baer algebra [10], so that $\mathbf{M}(G)$ is a type I operator algebra with Baer reduction.

2) Let G be a torsion free polycyclic-by-finite group. Then $\mathcal{F}_p[G]$ has no zero divisors [4] and no idempotents different from 0 and 1. Therefore $\mathcal{F}_p[G]$ is a Baer ring, and $\mathbf{M}(G)$ is a type I operator algebra with Baer reduction.

In the following three examples, $\mathbf{M}(G)$ is an operator algebra with the type III weak Baer reduction.

3) Let G be locally finite, prime, have no elements of order p , and all its conjugacy classes be countable. Then the ring $Q(\mathcal{F}_p[G])$ is simple and directly infinite [9].

Note that for a locally finite group G without order p elements, the ring $\mathcal{F}_p[G]$ is regular (see Theorem 1.5 in [15]). In this case, $Q(\mathcal{F}_p[G])$ is regular and right self-injective ([5], page 69); every such ring is a Baer ring [6]. For the present example, it follows from Corollary 9.21 in [6] that $Q(\mathcal{F}_p[G])$ is of type III.

4) If G is a solvable group containing a nontrivial locally finite normal subgroup, but no nontrivial finite normal subgroup, then $Q(\mathcal{F}_p[G])$ is a type III Baer ring. This result is a combination of Theorem 2.3 of [8], Proposition 7 from [7], and Theorem 2.10 from [15].

5) Let G be the group of permutations of an infinite set X moving only finitely many elements of X . Then $Q(\mathcal{F}_p[G])$ is a type III Baer ring; see Corollary 2.8 in [14].

The problem of constructing non-Archimedean operator algebras with type II Baer reductions remains open.

References

- [1] R. Baer, *Linear Algebra and Projective Geometry*, Academic Press, New York, 1952.
- [2] S. Bosch, U. G ntzner, and R. Remmert, *Non-Archimedean Analysis*, Springer, Berlin, 1984.

Figure 1: The open-problem statement and its immediate context in Kochubei, page 7 [3].

are uniformly bounded by one. Thus the ordinary Gauss norm on $K \otimes_k R$ is submultiplicative and its completion is the c_0 -space on E .

Left multiplication acts contractively on this completion. Because the basis contains 1, applying left multiplication by a to 1 recovers a and proves that its operator norm is exactly the Gauss norm. The regular representation is therefore a closed operator algebra. Finally, an element of the unit ball of $c_0(E, K)$ has only finitely many coefficients of norm one; reducing those coefficients gives an element of the original algebra R . The kernel consists exactly of the elements of norm strictly less than one. Consequently the operator-algebra reduction is R , with no enlargement caused by completion.

3 An equal-characteristic realization theorem

Theorem 1 (Realization theorem). *Let k be any field and let R be any unital associative k -algebra. There exist*

$$K = k((t)), \quad B = c_0(E, K), \quad \mathfrak{A} \subseteq \mathcal{L}(B),$$

where K has its t -adic absolute value, E is a Hamel basis of R over k , and $\mathfrak{A}$ is a norm-closed unital K -operator algebra, such that

$$\tilde{\mathfrak{A}} \cong R$$

as unital k -algebras.

Proof. Choose a Hamel basis E of R containing 1_R . On the algebraic scalar extension

$$A_{\text{alg}} = K \otimes_k R = \left\{ \sum_{e \in F} a_e e : F \subset E \text{ finite, } a_e \in K \right\}$$

put the Gauss norm

$$\left\| \sum_{e \in F} a_e e \right\|_G = \max_{e \in F} |a_e|.$$

For $e, f \in E$, write

$$ef = \sum_{g \in F(e,f)} c_{e,f}^g g, \quad c_{e,f}^g \in k,$$

with $F(e, f)$ finite. The t -adic absolute value is trivial on k , so $|c_{e,f}^g| \leq 1$. The ultrametric inequality therefore gives

$$\|xy\|_G \leq \|x\|_G \|y\|_G \quad (x, y \in A_{\text{alg}}). \quad (1)$$

The completion A of A_{alg} is, as a Banach K -space, canonically $c_0(E, K)$. Inequality (1) extends the multiplication uniquely to A and gives

$$\|xy\| \leq \|x\| \|y\| \quad (x, y \in A). \quad (2)$$

Thus A is a unital non-Archimedean Banach algebra and $\|1_R\| = 1$.

For $a \in A$, let $L_a \in \mathcal{L}(A)$ be left multiplication by a . Equation (2) gives $\|L_a\| \leq \|a\|$. The reverse inequality follows by evaluating at the unit:

$$\|L_a\| \geq \|L_a(1_R)\| = \|a\|.$$

Hence $a \mapsto L_a$ is an isometric unital algebra homomorphism. Its image

$$\mathfrak{A} = \{L_a : a \in A\} \subseteq \mathcal{L}(c_0(E, K))$$

is consequently norm closed.

It remains to identify the reduction. If $a = \sum_{e \in E} a_e e \in A$ and $\|a\| \leq 1$, then $(a_e)_{e \in E}$ is a c_0 -family in $\mathcal{O}_K$. In particular, only finitely many coefficients satisfy $|a_e| = 1$, so

$$\rho(a) = \sum_{e \in E} \overline{a_e} e \quad (3)$$

is a well-defined finite sum in R . Reduction of the structure constants shows that $\rho : A_1 \rightarrow R$ is a surjective unital algebra homomorphism.

We claim that $\ker \rho = A_0$. One inclusion is immediate. Conversely, if $\rho(a) = 0$, then $|a_e| < 1$ for every e . Since (a_e) is a c_0 -family, the set

$$F = \{e \in E : |a_e| \geq 1/2\}$$

is finite. If F is nonempty, the maximum of $|a_e|$ over F is strictly below one, while outside F all coefficients have norm below one half. Hence $\|a\| < 1$. If F is empty, directly $\|a\| \leq 1/2 < 1$. This proves the claim and yields $A_1/A_0 \cong R$.

The regular representation is isometric, so the unit balls and open unit balls of A correspond exactly to those of $\mathfrak{A}$. Hence $\tilde{\mathfrak{A}} \cong A_1/A_0 \cong R$, as required. $\square$

4 The type-II example

For a field k , let $\mathcal{M}_k$ denote von Neumann's algebraic continuous factor: the completion, in its unique rank metric, of

$$\varinjlim_n M_{2^n}(k), \quad x \mapsto \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix}.$$

Ara and Claramunt recall that $\mathcal{M}_k$ is a simple regular right-and-left self-injective ring and a continuous factor of type II_f [1, Introduction and Section 2]. A regular self-injective ring is Baer, and the type II_f classification says precisely that its unit is finite and it has no nonzero Abelian idempotent; this is type II in Kochubei's terminology [2, 3].

Corollary 2 (Type-II Baer reduction). *For every field k there is a complete nontrivially valued non-Archimedean field $K = k((t))$, a Banach space B over K with an orthonormal basis, and a norm-closed unital operator algebra $\mathfrak{A} \subseteq \mathcal{L}(B)$ such that $\tilde{\mathfrak{A}}$ is a type-II Baer ring. In fact one may arrange*

$$B = c_0(E, K), \quad \tilde{\mathfrak{A}} \cong \mathcal{M}_k.$$

Proof. Apply Theorem 1 to $R = \mathcal{M}_k$. □

This proves the requested existence statement. Notice that the construction is stronger at the realization level: the analytic step does not use any special feature of a continuous factor and can realize every unital k -algebra as a reduction.

5 Verification and limitations

The proof has four independently checkable points.

1. Every basis product has finite support, and its coefficients lie in the trivially valued subfield k ; this proves the Gauss-norm estimate.
2. Completion is $c_0(E, K)$, and the bounded multiplication extends uniquely.
3. The left regular representation is isometric because 1_R is a basis vector of norm one; hence the image is a closed operator algebra.
4. A c_0 -family has finite residue support, and the kernel argument after (3) proves that no extra residue elements appear.

No computational experiment is used as evidence. The only external structural input is the standard existence and type- II_f classification of $\mathcal{M}_k$.

The construction does not produce a group algebra. It also uses a coefficient field $k \hookrightarrow k((t))$ and therefore does not directly give an example over $\mathbb{C}_p$. The source paper's general definition allows arbitrary complete nontrivially valued K , while its final examples are presented under the Section 4 convention $K = \mathbb{C}_p$. Thus the packet's status is: candidate full solution for the literal field-unrestricted existence problem; candidate partial result if a reviewer reads the sentence as imposing the standing field $K = \mathbb{C}_p$.

Bounded novelty search. On August 9, 2026, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:1401.2268 and the phrases “type II Baer reduction”, “Baer reductions”, and “non-Archimedean operator algebra”. The local arXiv full-text corpus was searched for the same phrases. Web searches used the exact source title, the exact open-problem sentence, Kochubei’s name, and the same core phrases. These searches found the source paper and background work but no later paper claiming a type-II construction or an answer to this problem. Ara–Claramunt [1] supplies the algebraic continuous factor used here but does not identify the non-Archimedean realization or Kochubei’s question. Accordingly, novelty is plausible but not certified by an exhaustive bibliographic database search.

Human-review recommendation. The candidate should receive expert review. The main proof is short; the highest-value checks are the equality $\ker \rho = A_0$, compatibility of the two type-II conventions, and whether the source’s final sentence is meant field-unrestricted or specifically over $\mathbb{C}_p$.

References

- [1] P. Ara and J. Claramunt, *Uniqueness of the von Neumann continuous factor*, *Canad. J. Math.* **70** (2018), 961–982; arXiv:1705.04501.
- [2] K. R. Goodearl, *Von Neumann Regular Rings*, Pitman, London, 1979.
- [3] A. N. Kochubei, *Non-Archimedean group algebras with Baer reductions*, *Algebr. Represent. Theory* **17** (2014), 1861–1867; arXiv:1401.2268.